



**RAJIV GANDHI COLLEGE OF ENGINEERING AND  
TECHNOLOGY**

**COMPUTER SCIENCE AND ENGINEERING**

# ***ASSIGNMENT - 4***

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Teacher's Sign :

# Department Vision And Mission

## Vision

Cultivate student as technocrats, researchers and entrepreneurs in the field of computer science and engineering.

## Mission

- To impart quality education in computer science and Engineering, by means of learning techniques and value-added courses.
- To inculcate professional excellence and research culture by encouraging projects in cutting-edge technologies through industry interaction.
- To build leadership skills, ethical values and teamwork among the students.
- To strengthen the collaboration of department and industry through internships, mentorships and professional body activities.

## PART - A

1) List out the steps for mapping design to code?

- Identify classes
- Define properties
- Define methods
- Implement design patterns
- Consider Interfaces and Abstract classes
- Handle dependencies
- Test-First development
- Implement Association
- Implement Interaction diagrams
- Manage implementation

2) Differentiate access and view layer?

### Access Layer

The data access layer focus on managing data storage and retrieval providing back

### View Layer

The view layer primarily deals with the presentation and user interaction aspects of the application.

3) What do you mean by prototyping user interface?

User interface (UI) prototyping is an iterative analysis technique in which user are actively involved in the mocking up of the UI for a system. UI prototype have several purpose as an analysis artifact that enable you to explore the problem space with your stake holders.

4) Define persistence?

→ Persistence refers to the ability of data or objects to survive beyond the lifetime of the application that created them.

Essentially, it means data or objects can be stored and retrieved later, even after the application is ~~done~~ closed and restarted.

5) Differentiate between coupling and cohesion.

### Cohesion

Cohesion measures how closely related the elements within a single module (like a class)

### Coupling

Coupling measures the degree of interdependence between different modules

6) What is layered architecture?

→ Layered architecture is a design approach in computer science where the application is divided into separate layers that can be managed and maintained independently.

→ Each layer has a specific role in the application and can be tested and modified separately.

7) Give the rules to avoid bad design.

→ Focus on current requirements.

→ Avoid premature optimization.

→ Design pattern.

→ Cohesion and coupling.

→ Abstraction and Encapsulation.

8) Define Events, states and transition?

### Events

An event is a trigger or signal that causes a change from one state to another.

### States

A state is a collection or location that the system or object is in at a particular time.

### Transition

A transition is the movement or change from one state to another, triggered by a specific event.

- 9) Mention the task to be performed in access layer?
- The access layer, also known as the data access layer, is responsible for managing data storage and retrieval.
  - Its primary task is to provide a bridge between the business logic layer and the database or other data storage system.

- 10) Explain what is significance of Occam's razor?  
How does Occam's razor relate to o-o design?

→ Occam's Razor, a principle advocating for the simplest explanation when faced with multiple, equally valid options, is a cornerstone of good design and problem solving.

→ In o-o design, it encourages minimizing complexity by favoring simple, efficient solutions and eliminating unnecessary features or code.

## Part - B

11) Discuss in detail the steps involved in designing the access layer classes:

- The access layer (also known as the data Access layer or DAL) is responsible for all interactions with the data source, typically a relational or NOSQL database.
- It acts as a bridge between the business logic and the database by providing an abstraction for data operations.

→ Below are the detailed steps involved in designing this layer.

- + Analyze the data Requirements.
- + Identify the entities.
- + Define data transfer objects.
- + Design DAO interfaces.
- + Implement DAO classes.
- + Manage database connections.
- + Implement Exception handling.
- + Handle transactions.
- + Ensure Security and data validation.
- + Integrate with business logic layer.

### Analyze the data Requirements

→ Begin by understanding the data that the application needs to store, retrieve and manipulate.

→ This involves studying.

- + The database schema.
- + Entity relationships and dependencies.
- + Business process that involve data transactions.

| user  |
|-------|
| id    |
| name  |
| email |

| orders  |
|---------|
| id      |
| name    |
| prod-id |

| product |
|---------|
| id      |
| name    |
| price   |

### Identify the entities

- Entities represent real-world objects stored in the database. Each entity is typically mapped to a class in the access layer.
- These classes are often referred to as model or entity classes.

### Define data transfer objects.

- Data transfer objects are simple classes that carry data between layers.
- They contain attributes without any business logic.
- DTO's ensure that only the necessary data is stored access layer, maintaining encapsulation and separation of concerns

| user DTO |
|----------|
| id       |
| name     |
| email    |

## Design DAO Interface

→ DAO interfaces define the contract for data operations for each entity, you create an interface that specifies what operations can be performed each as.

- + create
- + Read
- + Update
- + Delete

→ These interface provide abstraction, allowing the implementation to be changed without affecting the rest of the application.

## Implement DAO classes.

The DAO interface or implemented in concrete classes, these classes contain the actual logic for interacting with the database, using technologies like:

- \* JDBC.
- + ORM frameworks like hibernate.
- + Entity Framework.
- + SQL Alechemy.

| user DAO Impl   |
|-----------------|
| get userById()  |
| get All users() |
| insert user()   |
| update user()   |
| delete user()   |

## Implement exception handling

- Data access operations can result in exceptions. The access layer should,
  - Catch and handle these exceptions.
  - Translate them into meaningful or custom exception.
  - Ensure that errors don't propagate up or unexpectedly to higher layers.

## Handle Transactions.

- For operations involving multiple data actions. Transaction management is essential to maintain data consistency and integrity. The access layer should,
  - + Begin a transaction,
  - + Commit it if all operations succeed,
  - + Roll back if any operations fails.

## Ensure security and data validation

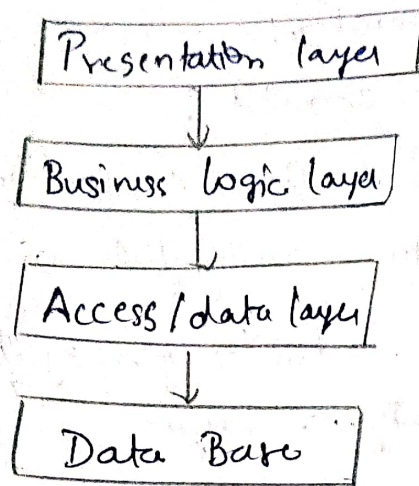
The access layer should include basic checks to prevent

- SQL Injection
- Unauthorized access to sensitive data.

## Integrate with Business Logic Layer

- Once the access layer classes are ready, they are integrated into the business logic layer.
- The business layer interacts with the access layer through DAO interfaces to perform data operations, ensuring a clean separation between

Logic and persistence.



12) Illustrate the purpose of view Layer interface using ATM banking system?

The view layer is the user interfaces part of an application it is responsible for:

- Interacting with the user.
- Displaying information.
- Accepting inputs
- Passing those input to the business logic for processing.

In an ATM system, the view layer is what the customer sees and interacts with for example.

- The ATM screen (GUI)
- Card reader interface
- Key pad for PIN entry.
- Touch screen buttons for transaction.

## Purpose of view layer interface in ATM

- Shows menu: withdraw, deposit, balance Inquiry.
- Accepts user choice, amount, PIN, etc.
- Displays message: Transaction successful, Incard PIN.
- Sends user input to business logic for processing.
- Forms and data entry withdraw.
- Applications windows.

## Example Use case:

- User inserts card - view display "Enter PIN".
- User enters PIN -> now passes it to logic for verification.
- View shows options: withdraw, deposit, balance Inquiry.
- User selects withdraw and enters amount.
- View sends request to logic layer.
- After processing view displays "Take your cash" and prints receipt.

## View layer diagram for ATM system

The view layer interface in an ATM banking system:

- Acts as a bridge between the user and the system.
- Acts as a bridge between the user and the systems.
- Ensures that users can perform operations easily.
- Displays output in a user friendly way.
- Captures input and forwards them for processing.

